

Evaluating the causes of cost reduction in photovoltaic modules

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ABSTRACT

Photovoltaic (PV) module costs have declined rapidly over forty years but the reasons remain elusive. Here we advance a conceptual framework and quantitative method for quantifying the causes of cost changes in a technology, and apply it to PV modules. Our method begins with a cost model that breaks down cost into variables that changed over time. Cost change equations are then derived to quantify each variable's contribution. We distinguish between changes observed in variables of the cost model – which we term low-level mechanisms of cost reduction – and research and development, learning-by-doing, and scale economies, which we refer to as high-level mechanisms. We find that increased module efficiency was the leading low-level cause of cost reduction in 1980–2012, contributing almost 25% of the decline. Government-funded and private R&D was the most important high-level mechanism over this period. After 2001, however, scale economies became a more significant cause of cost reduction, approaching R&D in importance. Policies that stimulate market growth have played a key role in enabling PV's cost reduction, through privately-funded R&D and scale economies, and to a lesser extent learning-by-doing. The method presented here can be adapted to retrospectively or prospectively study many technologies, and performance metrics besides cost.

1. Introduction

Photovoltaics have exhibited the most rapid cost decline among energy technologies (Trancik and Cross-Call, 2013) (Fig. 1). In parallel with cost declines and performance improvement, global PV deployment has grown rapidly (Trancik, 2014). Continued PV deployment could help reduce greenhouse gas emissions and other pollution from energy systems (Hertwich et al., 2015), and contribute to climate change mitigation (Trancik and Cross-Call, 2013). For PV deployment to experience sustained growth in the future, however, particularly when considering the additional costs of addressing solar intermittency (Braff et al., 2016), further cost declines are likely needed (U.S. Department of Energy, 2012). This paper aims to identify the causes of PV's rapid cost declines in the past and gain insight into maintaining the pace of improvement in the future. More fundamentally, we aim to advance a model for understanding the mechanisms of technology improvement at multiple levels, from human efforts to devices, that can be applied to many technologies and measures of performance.

Improvement trends in PV and other technologies have been studied by various research communities. Correlational analysis is a common approach in these studies, often focusing on cost (or other measures of performance) and production or research investment levels (Nagy et al., 2013). One of the most widely-used models is the experience curve,

which relates a technology's cost to cumulative production as a power law. Using this relationship as an explanatory or predictive tool, studies have estimated the rates of performance improvement for a range of technologies (Grubler et al., 1999; Koh and Magee, 2008; Nagy et al., 2013; Rubin et al., 2015; Zheng and Kammen, 2014). For example, PV module costs fell by about 20% with every doubling of cumulative capacity since the 1970s (McDonald and Schrattenholzer, 2001; Nemet, 2006). Several explanations for this cost decline have been proposed, such as public research and development efforts and various consequences of market growth (Bettencourt et al., 2013), including learning-by-doing, economies of scale, and private research and development efforts (Funk, 2013; McDonald and Schrattenholzer, 2001; Sagar and van der Zwaan, 2006; van der Zwaan and Rabl, 2004; Yu et al., 2011; Pillai, 2015). These studies share an approach to examining technology cost evolution where important high-level drivers of cost reduction are assumed and their influence on cost is inferred based on correlation. Technologies are treated as black boxes and the causes of cost reduction within a technology are not modeled mechanistically.

Another group of studies uses detailed, device-level cost models, to understand how features of a technology or manufacturing process contribute to costs at one or more snapshots in time. Several such studies exist for PV, and they provide information on how individual cost components contribute to total costs, while taking into account the physics of PV

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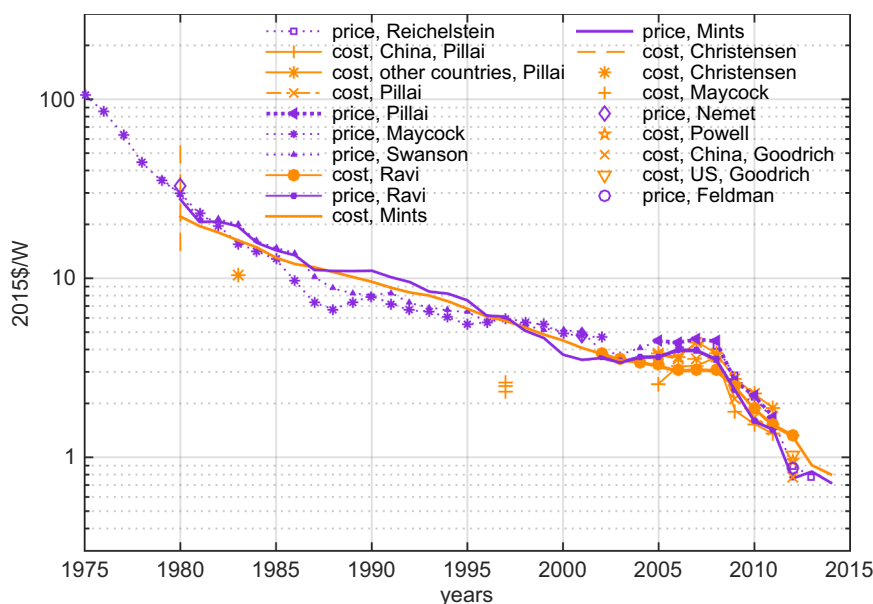


Fig. 1. Module costs and prices since 1975. Costs are shown in orange, and prices are shown in purple. References: (Reichelstein and Sahoo, 2014); (Pillai and Cruz, 2013); Maycock price data from (Nemet, 2006) and cost data from (Maycock, 1997); (Swanson, 2011); (Ravi, 2013); (Mints, 2015); (Christensen, 1985); (Nemet, 2006); (Powell et al., 2013); (Goodrich et al., 2013b); (Feldman et al., 2014). Values are averages across different crystalline silicon PV technologies except for those in (Powell et al., 2013) (multicrystalline silicon) and (Goodrich et al., 2013b) (monocrystalline silicon).

technologies (Goodrich et al., 2013a; Powell et al., 2012, 2013; Woodhouse et al., 2013a, 2013b). They also propose avenues for future technical improvement at the device or manufacturing level, and estimate cost reductions that might be achieved in the future (Jones-Albertus et al., 2016). Missing from these studies, however, is a method of accurately quantifying how each change to a feature of the technology or manufacturing process contributes to cost reductions, when many changes occur simultaneously. This knowledge is needed to understand the mechanisms of cost reduction but requires further modeling advances.

Pursuing both dynamic and detailed, device-level models is critical for identifying the causes of improvement in PV and other technologies. This combined approach would address inherent limitations in using correlational analyses to identify causal effects. This approach would also address the lack of dynamics in device-level studies. A few past studies have begun to develop such a methodology by decomposing technology costs over time (McNerney et al., 2011; Nemet, 2006). A study of the drivers of PV module cost changes from the 1970s to the early 2000s (Nemet, 2006) pioneered a bridge of this kind, and found that learning-by-doing had a limited effect on cost reductions.

In this paper we propose a new conceptual framework and dynamic-yet-detailed quantitative model for analyzing PV's (or any technology's) cost evolution. We start with a cost equation that computes costs from a set of variables, such as module efficiency, wafer area, and manufacturing plant size. From this we derive cost change equations that estimate the contribution of each variable to cost changes. Multiple simultaneous changes to variables have different impacts on cost than individual changes summed together, and this must be accounted for in attributing cost changes to individual variables. Our method of estimating variable contributions is derived from adapting the total differential of cost (which decomposes infinitesimal cost changes) to finite changes.

In attributing PV's cost decline to particular causes, we draw a distinction between *low-level* causes (or mechanisms) and *high-level* causes (or mechanisms). Low-level mechanisms explain cost reduction in terms of changes to variables of a cost model, representing measurable and technology-specific determinants of cost (e.g. wafer area). High-level mechanisms explain cost reduction in terms of processes like R&D, learning-by-doing, and scale economies that subsume low-level cost reductions. Both low- and high-level mechanisms can simultaneously provide explanations for a technology's cost change. For example, suppose a technology realizes an improvement to yield from learning-by-doing on the factory floor. The resulting cost reduction can be explained in two ways. One way is to say that yield increased; the other way is to say that learning-by-doing drove down costs. Both explanations are correct, and emphasize different views of

the process of improvement. The explanation based on yield improvement (a low-level mechanism) ties the cost reduction to the detailed, device-level cost model of this technology. The explanation based on learning-by-doing (a high-level mechanism) ties the cost reduction to a general improvement mechanism that is discussed widely in studies of historical technology evolution. Here we consider both levels, and thus bridge bottom-up and top-down approaches to understanding technology cost evolution.

By considering both the low-level and high-level causes of PV's improvement, we uncover lessons that are useful for a variety of decision-makers. These may include engineers who design and manufacture PV modules, or firm managers and government policy-makers who develop strategy to support technological development. For example, our findings contribute to a long-standing debate concerning the effect of public investments in R&D versus market-expansion policies (Duke and Kammen, 1999; Hoppmann et al., 2013; Zheng and Kammen, 2014).

We focus on crystalline silicon PV modules because of their long history and dominant market share among PV technologies (Fraunhofer Institute, 2017). A key goal of our analysis is to understand the mechanisms of PV technology improvement and cost reduction over time, making it essential to study costs over a long time period. Since the 1950s, this technology has improved steadily due to R&D and manufacturing efforts (Powell et al., 2012). We analyze the costs starting in 1980, when space applications of PV were overtaken by terrestrial applications, which did not require as high quality and reliability (Candelise et al., 2013; Green, 2005; Nemet, 2006). We look at typical costs globally, since PV modules are manufactured and traded globally. We focus on costs rather than prices because mechanisms of technology improvement are reflected directly in costs, while prices also include mark-ups that are influenced by other factors, such as market competition (Pillai and McLaughlin, 2013). The method we develop can be adapted to study PV systems as a whole (including non-module cost components that show significant potential for cost reduction (Fraunhofer Institute, 2015; Trancik et al., 2015)), and a wide range of other technologies and measures of performance other than cost (Carbajales-Dale et al., 2014; Hertwich et al., 2015; Needell et al., 2016). The method might also prove a useful quantitative framework for eliciting high-quality input from experts on the prospects for future technological improvements (Morgan, 2014).

This paper is organized as follows: Section 2 provides a detailed explanation of the cost model. Section 3 explains the method of attributing cost changes to variables. Section 4 shows the results of our analysis and the connection between low-level and high-level mechanisms. In Section 5 we discuss the implications for future developments in PV and conclude.

2. Cost model

2.1. Cost decomposition strategy

We first develop a cost model for PV modules. The cost components are calculated based on quantities (or usage ratios) ϕ and prices of inputs p used in manufacturing.

$$C_m \left(\frac{\$}{\text{module}} \right) = \underbrace{\frac{1}{y_m} \sum_{i \neq c, w} \phi_{mi} p_i}_{\text{non-cell module costs}} + \underbrace{\frac{n_{mc}}{y_m y_c} \sum_{i \neq w} \phi_{ci} p_i}_{\text{non-wafer cell costs}} + \underbrace{\frac{n_{mc} n_{cw}}{y_m y_c y_w} \sum_i \phi_{wi} p_i}_{\text{wafer costs}} \quad (1)$$

where

y_m	yield at module manufacturing
y_c	yield at cell manufacturing
y_w	yield at wafer manufacturing
ϕ_{mi}	quantity of input i per module
ϕ_{ci}	quantity of input i per cell
ϕ_{wi}	quantity of input i per wafer
p_i	price of input i
n_{mc}	number of cells per module
n_{cw}	number of wafers per cell.

While this equation has been written to represent wafer, cell, and module costs, which is a decomposition scheme specific to PV, the formulation of costs in terms of usage ratios (ϕ) and input prices (p) is a general one that can describe any technology. ϕ variables generally change as the result of engineering efforts to improve efficiency and materials utilization, while p variables change due to bulk purchasing, scarcity or other market effects (Kavlak et al., 2015), or input substitutions.

At each of the three levels of PV manufacturing costs – wafer production, cell production, and module production – there are costs for materials, labor, operation & maintenance, electricity, and depreciation of the plant and equipment (Fig. 2). Decomposing by module, cell, and wafer levels disaggregates the production process, but creates challenges here for estimating the sources of cost reduction over time. A consistent categorization of costs is needed for every time period of interest starting with 1980, but such early cost data is scarce. Instead we accomplish this consistency over time by decomposing module production costs into three components by input type: silicon costs, non-silicon material costs, and plant size-dependent costs. These components are further modeled as described below.

2.2. Silicon costs

Historical prices of silicon (i.e. polysilicon) can be obtained from the literature (de La Tour et al., 2013; Nemet, 2006; Yu et al., 2011) or from

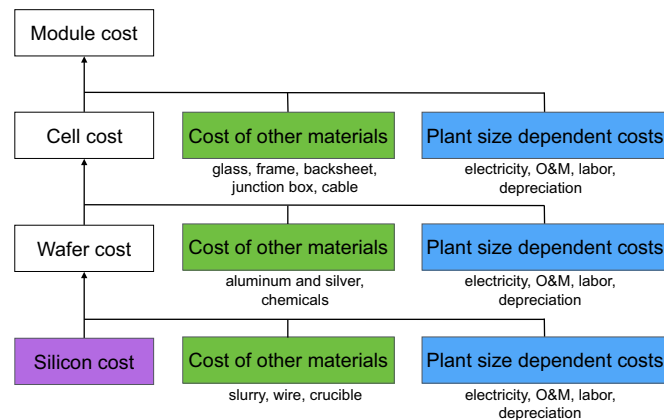


Fig. 2. Cost components for wafer, cell, and module levels (vertical disaggregation) and different input types (horizontal disaggregation).

Table 1

Data used to calculate module cost components. Table S1 provides the individual data collected from the references below and the approach for obtaining the central estimate of each variable.

Factor	Unit	1980	2001	2012	References and notes
Plant size (K)	MW/yr	1	13.3	1000	1980: (Swanson, 2006), (Mints, 2015). 2001: (Mitchell et al., 2002), (Mints, 2015), 2012: (Pillai, 2015), (Powell et al., 2012), (Mints, 2015).
	modules/yr	1.72×10^4	8.09×10^4	3.35×10^6	Calculated by dividing the plant size in MW/yr by module power output in each year. We use the modules/year values to populate the plant size variable in our model.
Module efficiency (η)	unitless	8%	13%	15.2%	1980: (Mints, 2015), (Christensen, 1985), (Maycock, 1995). 2001: (Mints, 2015), (Rohatgi, 2003), (Symko-Davies et al., 2000). 2012: (Mints, 2015), (Powell et al., 2012).
Polysilicon price (p_s)	2015\$/kg	126	36	26	1980: (Mints, 2015), (Williams, 1980), (Costello and Rappaport, 1980), (Christensen, 1985). 2001: (Mints, 2015), (Swanson, 2006). 2012: (Mints, 2015), (Powell et al., 2013).
Wafer area (A)	cm ²	90	156	243	1980: (Christensen, 1985), (Mints, 2015). 2001: (Mints, 2015), (Swanson, 2006). 2012: (Powell et al., 2012), (Mints, 2015).
Silicon thickness (t)	μm	500	300	180	1980: (Swanson, 2006), (Fraunhofer Institute, 2017). 2001: (Swanson, 2006), (Sarti and Einhaus, 2002), (Green, 2005), (Fraunhofer Institute, 2017). 2012: (Applied Materials, 2011), (Fraunhofer Institute, 2017), (Goodrich et al., 2013a).
Silicon utilization (U)	unitless	0.20	0.36	0.45	1980: (Hammond, 1977). 2001: (Sarti and Einhaus, 2002). 2012: (Centrotherm Photovoltaics, 2010).
Silicon usage (v) $\equiv t/U$	cm	0.25	0.083	0.04	See the references for t and U .
Yield (y)	unitless	75%	86%	95%	1980: (Mints, 2015). 2001: (Rohatgi, 2003), (Mints, 2015), (Sarti and Einhaus, 2002), (Little and Nowlan, 1997). 2012: (Applied Materials, 2011), (ITRPV, 2012), (Mints, 2015).
Share of materials costs (θ)	unitless	0.69	0.43	0.65	1980: (Williams, 1980). 2001: (Maycock, 1997). 2012: (Powell et al., 2013).
Scaling factor (b)	unitless	0.27	0.27	0.27	(Maycock, 1997).
Module cost	2015\$/W	29.07	4.08	1.08	1980: (Mints, 2015), (Christensen, 1985). 2001: (Mints, 2015), (Goodrich et al., 2013b), (Powell et al., 2013), (Ravi, 2013).

industry sources (Mints, 2015). The amount of silicon used per wafer is a function of wafer area, silicon density, silicon layer thickness, and silicon utilization (the fraction of the silicon ingot used in the wafer after accounting for losses). Multiplying by the number of cells and the price of silicon, total silicon cost for the module can be expressed as

$$\text{Si cost} = n_{mc} \frac{Ah\rho}{U} p_s = n_{mc} A v \rho p_s. \quad (2)$$

Here n_{mc} is the number of cells per module, A is wafer area, h is wafer thickness, $\rho = 2.33 \text{ g/cm}^3$ is wafer density, U is silicon utilization, and p_s is the price of polysilicon. We define the combination $v \equiv h/U$, which we refer to as ‘silicon usage’ for simplicity. The data for these variables are provided in Table 1.

2.3. Non-silicon materials costs

Non-silicon materials include the crucible used to produce silicon ingots; slurry and wire used for wire-sawing; aluminum and silver pastes, chemicals and screens used in cell manufacturing; and glass, frame, backsheets, encapsulant, ribbon, junction box and cable used in the module (Powell et al., 2013). To a first approximation, the usage of these materials can be categorized as proportional to wafer area (e.g. aluminum pastes), proportional to module area (e.g. glass), proportional to module perimeter (e.g. frame), or none of the above (e.g. junction box), so that costs would take the form

$$\text{non-Si materials costs} = c_0 + n_{mc} c_1 A + c_2 A_m + c_3 P \quad (3)$$

with A representing wafer area, A_m the module area, P the module perimeter, and the c_i various constants. Because of data limitations, and since most materials costs depend on area, we ignore the fixed and perimeter-dependent categories in this expression. Since the late 1970s wafer area A and module area A_m have increased proportionally,¹ $A \propto A_m$. Thus we simplify Eq. (3) to

$$\text{non-Si materials costs} = n_{mc} c A, \quad (4)$$

where c is the per-area cost of all non-silicon materials and A is wafer area. Here the non-Si materials costs account for the costs at all of the wafer, cell, and module levels. We derive the value of c from estimated materials costs in the three time periods. Based on the literature, the share of materials costs in PV modules varied between 43% and 69% (Williams, 1980; Maycock, 1997; Powell et al., 2013). We calculate the materials costs using the total module cost and the fraction due to materials, subtract the cost of silicon, and divide out the wafer area to obtain c .

2.4. Plant size-dependent costs

We model electricity, labor, maintenance, and depreciation costs per wafer as a group of costs that are expected to experience scale economies with increases in PV plant scale (Maycock, 1997). In many manufacturing processes it is common for both capital and operating costs to rise less-than-proportionately with plant output. Plant size-dependent scale economies occur for many reasons, including greater amortization of fixed costs and machine scaling effects (Silberston, 1972). Production efficiencies may also occur, such as with labor tasks that become more specialized in larger plants, leading to more efficient workers (Silberston, 1972). In practice it is difficult to separate these many effects, and here we

¹ The relationship between wafer area and module area is given by

$$A_{\text{module}} = \frac{n_{mc} A}{\alpha}$$

where n_{mc} is the number of cells per module and α is the area utilization, the fraction of module area used by cells. Both wafer area and module area increased about three-fold between 1980 and 2012, while α and n_{mc} have stayed almost constant in a typical module (Christensen, 1985; Powell et al., 2013).

capture them as a whole using a power-law model based on PV plant scale,

$$\text{plant size-dependent costs} = n_{mc} p_0 \left(\frac{K}{K_0} \right)^{-b}, \quad (5)$$

where K_0 is a reference plant size, p_0 represents the total of these costs for a plant with the reference size, and b is the scaling factor. For convenience we take K_0 as the number of modules manufactured in a year in a 1000 MW plant (the typical plant size value for 2012), though this choice is just a convention since the effects of a different choice of K_0 would be absorbed into a different value for p_0 . We use $b = 0.27$ as the scaling factor (Maycock, 1997). We obtain this value by computing the change in electricity, labor, maintenance, and depreciation costs between plants of two sizes described in Maycock (1997). We obtain p_0 for 2012 from Powell et al. (2013). For 1980 and 2001 we compute p_0 by computing non-materials costs and dividing out the factor $(K/K_0)^{-b}$.

2.5. Final cost equation

The power output of a module M is given by

$$M = \frac{\sigma n_{mc} A \eta}{\alpha} \quad (6)$$

where $\sigma = 0.1 \text{ W/cm}^2$ is the solar constant, n_{mc} is the number of cells per module, A is wafer area, η is module efficiency, and α is module area utilization. We assume a constant value of $n_{mc} = 72$. Some wafers, cells, and modules created during production are faulty and must be discarded, leading to waste and additional costs. To account for this we include the production yield y .² Summing the three components of module costs, and dividing by module capacity, total costs are

$$C \left(\frac{\$}{W} \right) = \frac{\alpha}{\sigma A \eta y} \left[A v \rho p_s + c A + p_0 \left(\frac{K}{K_0} \right)^{-b} \right]. \quad (7)$$

Developing the cost model means choosing a set of variables to include, and the level of detail to expose about PV modules. The choice of variables used here reflects data availability, as well as what aspects of PV modules have received focus by researchers and industry. In principle, variables in the model above could be decomposed further to produce a model with additional detail. For example, module efficiency could be disaggregated into other variables, such as open-circuit voltage, short-circuit current, and fill factor, which in turn could be further disaggregated. Efficiency, however, is a major cost-determining variable, and its cost contributions are important regardless of whether they came through open-circuit voltage, fill factor, or another variable. Also, theoretically one could model all of the cost components including the materials, electricity, labor, and so on as dependent on both plant size and wafer area. However, the data to populate such a sophisticated model is not available. Therefore we make a compromise and model the electricity, labor, maintenance, and depreciation costs as scaling with plant size, while we model materials costs as dependent on wafer area.

We populate Eq. (7) with historical data from three snapshots in time (1980, 2001, and 2012) (Table 1) and obtain the three cost components for 1980, 2001, and 2012. The cost components are illustrated in Fig. 2 and their values are shown in Table 2. While all of the cost components have gone down in units of \$/W, their shares of total cost have varied. In particular silicon has become a smaller fraction of total cost over time, while non-silicon materials have become a larger fraction. The share of the plant size-dependent costs increased between 1980 and 2001 and then decreased after 2001.

Our decomposition variables are similar to Nemet (2006) (and one of the time periods we consider (1980–2001) is the same as in Nemet, (2006)), though there are a few important differences which are

² Wafer, cell, and module production each have individual process yields, though for simplicity we represent overall yield with one value.

Table 2

Cost components in 1980, 2001, and 2012. Costs are in 2015 US dollars.

Cost component	1980		2001		2012	
	\$/W	%	\$/W	%	\$/W	%
Silicon cost	10.88	37%	0.55	13%	0.15	14%
Non-silicon materials cost	9.17	32%	1.21	30%	0.56	51%
Plant size-dependent cost	9.01	31%	2.33	57%	0.38	35%
Total module cost	29.07	100%	4.08	100%	1.08	100%

summarized here. We include the contribution of non-silicon material costs, which are not considered in Nemet (2006). To avoid double-counting the reductions from efficiency improvements, we model silicon usage in units of g/module instead of g/W. Similarly we model plant output in units of modules/year instead of W/year. The share of costs that we find to be plant size-dependent and area-dependent are different. In Nemet (2006), all costs are modeled as being plant size-dependent while we model only non-materials costs as plant size-dependent, based on Maycock (1997). Also based on Maycock (1997) we use a higher exponent for scale-dependent costs, $b = 0.27$ versus $b = 0.18$ in Nemet (2006). Therefore, compared with Nemet (2006) our plant scale variable affects a smaller fraction of total module costs, while influencing this fraction more strongly. We similarly model all materials costs as area-dependent, and all non-materials costs as non-area-dependent. As a result a much larger fraction of costs are independent of wafer area in our model, 31–57% depending on the period versus 4% in Nemet (2006). Other differences in the modeling approach used here and in Nemet (2006) are discussed at the end of Section 3.

Our approach also differs from the one taken in Pillai (2015), which uses a cost model which doesn't distinguish between low-level technical variables (e.g. module efficiency) and high-level effects (e.g. industry investment). First, we include the effects of other materials costs in addition to silicon in our model. More fundamentally, our approach focuses on mechanistic relationships between cost components and variables, captured in detailed, device-level cost models, in order to study variables with a direct effect on cost components. For many technologies, there are known engineering relationships that provide substantial information about the causes of cost change. These relationships constrain what variables enter the cost model and the functional forms they appear in. For high-level mechanisms such as R&D investment to drive down costs, they must affect one or more of these underlying variables. Separating low and high in our analysis thus allows mechanisms at different levels to be non-competing explanations of cost reduction. For example, industry R&D investment and lower silicon usage may be simultaneously correct explanations for some portion of cost reduction, to the extent that the R&D is what led to lower silicon usage. Moreover, the low-level mechanisms identified in our study provide useful insight on how high-level mechanisms affect cost. For example, knowing that R&D led to cost reduction through a particular set of low-level mechanisms (e.g. conversion efficiency and others) can help with identifying potential future constraints or opportunities for R&D policy.

3. Attributing cost changes to variables

How much of the cost reduction in PV modules came from each of the variables in Eq. (7)? While it is simple to compute how much Eq. (7) changes when its variables change, it is more difficult to say what any individual variable contributed. Here we describe a general method for decomposing cost changes into contributions from individual variables. We outline the approach here and give the full derivation in Supporting Information Section S2. For any function $C(r_1, r_2, \dots)$, an infinitesimal change dC can be straightforwardly split into contributions from individual variables using the total differential,

$$dC = \sum_z \frac{\partial C}{\partial r_z} dr_z = \sum_z dC_r. \quad (8)$$

Here a summation term $dC_r = (\partial C / \partial r_z) dr_z$ accounts for the impact on C of a small change in variable r_z . Over a period of time t_1 to t_2 , this term contributes a finite change to C given by the integral

$$\Delta C_z(t_1, t_2) = \int_{t_1}^{t_2} \left(\frac{\partial C}{\partial r_z} \right) \frac{dr_z}{ds} ds. \quad (9)$$

Eq. (9) gives the contribution of variable z to changes in C between t_1 and t_2 . Summing over the contributions from all variables then recovers the total change, $\Delta C(t_1, t_2)$.

The challenge is to approximate this integral in a practical setting, where data about the technology's cost and variables have been sampled at discrete times. To do this, we first note that a cost can always be decomposed into a sum over cost components i , $C(\mathbf{r}) = \sum_i C_i(\mathbf{r})$, which typically account for different categories of inputs. In addition, the cost components are often (as in Eq. (7)) products of functions of variables,

$$C_i(\mathbf{r}) = C_i^0 \prod_z g_{iz}(r_z), \quad (10)$$

where C_i^0 is a constant and each function $g_{iz}(\cdot)$ gives the dependence of cost component i on variable z . Then approximating the integral in Eq. (9), it can be shown that the contribution of the z th variable to the change in total cost from time t_1 to time t_2 is

$$\Delta C_z(t_1, t_2) \approx \sum_i \tilde{C}_i \ln \left(\frac{g_{iz}(r_z^2)}{g_{iz}(r_z^1)} \right), \quad (11)$$

where r_z^1 and r_z^2 are the values of r_z at t_1 and t_2 , and $\tilde{C}_i$ is a representative value of cost component i during the period. There are several ways the representative value $\tilde{C}_i$ could be computed; in the SI, we show that a particularly good choice is given by $\tilde{C}_i = (C_i^2 - C_i^1) / (\ln C_i^2 - \ln C_i^1)$, where C_i^1 and C_i^2 are the values of cost component i at t_1 and t_2 .³ With this choice the variable contributions computed using Eq. (11) sum to the total change in the cost of the technology, $\Delta C = \sum_z \Delta C_z$. Eq. (11) thus produces a decomposition of ΔC , adapting the decomposition of dC in Eq. (8) to finite changes in cost. The method here is then a two-step procedure. First, one writes down (through knowledge of a technology) an equation for cost C as a function of a set of variables $\mathbf{r}$. Then, between two times for which one has data on the variables, one computes Eq. (11) for each variable.

In our case, the variables are $\mathbf{r} = (A, \eta, y, K, p_s, v, c, p_0)$. The three cost components in Eq. (7) can be written in the form of Eq. (10) as

$$C_1(\mathbf{r}) = \left(\frac{\alpha \rho}{\sigma} \right) v p_s \eta^{-1} y^{-1} \quad (12)$$

$$C_2(\mathbf{r}) = \left(\frac{\alpha}{\sigma} \right) c \eta^{-1} y^{-1} \quad (13)$$

$$C_3(\mathbf{r}) = \left(\frac{\alpha}{\sigma K_0^{-b}} \right) p_0 K^{-b} A^{-1} \eta^{-1} y^{-1}, \quad (14)$$

which from top to bottom are silicon costs, non-silicon materials costs, and plant size-dependent costs. Each cost component is a constant pre-factor multiplied by a product of functions of the variables. Eq. (11) can then be computed to give estimates for individual variables. For example, the cost change due to a change in efficiency is

$$\Delta C_\eta = \sum_{i=1}^3 \tilde{C}_i \ln \left(\frac{(\eta^2)^{-1}}{(\eta^1)^{-1}} \right) = - \sum_{i=1}^3 \tilde{C}_i \ln \left(\frac{\eta^2}{\eta^1} \right). \quad (15)$$

The values of $\tilde{C}_i$ can be computed from the cost components at the beginning and end of each period being studied. As an example, cost component 1 changed from 10.88 \$/W in 1980 to 0.55 \$/W in 2001 (Table 2),

³ In the SI we also show several other properties of $\tilde{C}_i$. It is related to and larger than the geometric mean of cost component i in the two periods, $C_i^{\text{geo}} = \sqrt{C_i^1 C_i^2}$, and a good approximation in practice is a value about 1/3 of the way from the geometric mean to the arithmetic mean, $\tilde{C}_i = \frac{1}{2}(C_i^1 + C_i^2)$.

leading to a value for $\tilde{C}_1$ of 3.46 \$/W.

An important departure from the method of Nemet (2006) is that we start with a cost equation first rather than beginning directly with cost change equations. This two-step approach has significant advantages. It ensures that our cost change equations are consistent with a realizable cost model, whose values can be directly compared with actual costs. A cost model shows explicitly how variables jointly determine total cost, making it easier to see what modeling assumptions are being made. It also helps to avoid double-counting or undercounting the effects of variables on costs, because the dependence of cost on each variable has been fully accounted for in the cost model.⁴

4. Results and discussion

In this section we first discuss the *low-level mechanisms* of module cost reduction, which refer to changes to variables in the cost equation. We then relate the low-level mechanisms to *high-level mechanisms* of cost reduction, which refer to strategies (e.g. R&D investment) or emergent effects (e.g. learning-by-doing, economies of scale) that drive down costs by affecting one or more low-level mechanisms.

4.1. Low-level mechanisms of cost reduction

Fig. 3 and Table 3 show the changes in module cost due to each variable in the two periods, 1980–2001 and 2001–2012, and the entire period, 1980–2012. Improving efficiency was the largest contributor in the first period, responsible for 24% of the cost reduction. In the second period, module efficiency was only the fourth most significant factor, and its contribution dropped to 12%. Efficiency increased through many improvements, such as surface passivation (Green, 2005), anti-reflective coating (Goetzberger et al., 2003), texturing of the wafers (Bruton, 2002; Rohatgi, 2003), and development of encapsulant materials (Green, 2005).

The second most significant factor in 1980–2001 was lower per-wafer area costs of non-silicon materials, contributing 22% of the cost decline. The contribution of this factor in 2001–2012 was 15%. Non-silicon materials include substances such as glass, laminate, and metal paste that become embedded in the module as well as slurry and wire used during production. The cost of these materials has been reduced by various process and module design improvements. For example, the cost of slurry used in wafer cutting may be reduced by recycling, with recycling rates up to 80% reported (Applied Materials, 2011).

Changes in polysilicon price contributed about 18% and 3% of the module cost decline in 1980–2001 and 2001–2012, respectively. The endpoints of our analysis lie on either side of a temporary period of silicon shortage in 2005–2008, during which polysilicon prices surged. Before this period, most polysilicon was used by the semiconductor industry. The PV industry used electronic grade silicon rejects from the semiconductor industry, which has higher purity requirements (Sarti and Einhaus, 2002). Around 2006, polysilicon demand by the PV industry surpassed that of the semiconductor industry (Fisher et al. 2012). While more than 80% of the global polysilicon production was consumed by the semiconductor

industry before 2000 (Fu et al., 2015), about 90% went to manufacturing PV cells as of 2012 (U.S. Geological Survey, 2012).

Decreasing silicon usage contributed 15% to module cost reduction in 1980–2001 and 8% in 2001–2012. Silicon usage depends on wafer thickness h and silicon utilization U . To study the total cost reduction from both variables, we defined the combination $v \equiv h/U$. In 1980–2001 wafer thickness decreased from 500 μm to 300 μm while silicon utilization increased from 20% to about 35%. Reduced thickness and higher utilization contributed about equally to the silicon cost reduction in this period. Silicon usage continued to decrease in 2001–2012, though it had a less significant cost impact. The industry developed thinner wafers both to reduce the cost of silicon and to increase conversion efficiency (Goetzberger et al., 2003). Silicon utilization also increased, though losses remain high, with about 50% of entering silicon lost during slicing of wafers from silicon ingots. Decreasing thickness contributed about 70% of the silicon-usage-related cost reduction in this period while increasing utilization contributed the remaining 30%.

Increasing wafer area also contributed to module cost reduction, increasing from 90 cm^2 to 156 cm^2 in 1980–2001, and growing significantly again to about 240 cm^2 by 2012. Increasing wafer area, given a fixed number of cells per module, means that each module assembled produces more power. Material costs are mostly proportional to area, but other assembly costs are insensitive to area (Ghannam et al., 1997; Nemet, 2006), so that larger wafer area leads to cost savings.

Process improvements led to increasing yields in wafer, cell, and module production. Overall yield increased from 75% in 1980 to 95% in 2012. The change in yield contributed about 7% to the module cost decline in both 1980–2001 and 2001–2012. Reduced handling of wafers, cells, and modules due to automation, and improvements in processes such as wafering, helped to increase yield (Applied Materials, 2011). We note that other improvements (such as larger wafer sizes) can decrease yields (Bruton, 2002), so that yield considerations can be a limiting factor for otherwise cost-saving practices.

The pre-factor p_0 in Eq. (5) provides the level of plant size-dependent costs for a plant of a fixed size K_0 , thus accounting for the overall level of electricity, labor, maintenance, and depreciation costs at each time. The change in p_0 was estimated to have increased cost in the 1980–2001 period and decreased cost in the 2001–2012 period. As described in Section 2 we calibrate p_0 in 1980 and 2001 by requiring non-material costs in our cost model to match values from our data. We regard the variable with caution since changes are difficult to interpret and it is likely to propagate uncertainty in the data. However, its effects are among the smallest in both periods.

Finally, increasing manufacturing plant sizes resulted in scale economies through shared infrastructure, reduced labor requirements, higher yield, and better quality control (del Canizo et al., 2009). Typical plant sizes have scaled up with the industry, starting from 1 MW in 1980 and growing to about 13 MW in 2001 and to 1000 MW in 2012. Plant size became an especially significant factor in the more recent period, contributing almost 40% of the decline in module cost.

In Section S3 of the Supporting Information we estimate the sensitivity of our results to the uncertainty in the input variables shown in Table 1. We conclude that overall our results are robust to changes in the variables.

Our findings show some similarities to earlier reported results, as well as some differences. Similar to Nemet (2006) we find that increasing module efficiency was the largest contribution to cost reduction in 1980–2001. While Nemet (2006) finds the sources of cost declines to be heavily concentrated in plant size and module efficiency changes, we find cost-reducing effects were spread across a number of variables. In Nemet (2006) the costs of non-silicon materials are not considered, though we find that non-silicon material costs contributed almost as much to cost reduction as silicon did in this period. We obtain a lower estimate for the contribution of plant size in this period (8% versus 43%), and larger contributions for wafer area (11% versus 3%), silicon consumption (15% versus 3%), yield (7% versus 2%), and silicon price (18% versus 12%). Comparing our results from the 2001–2012 period with Pillai (2015), which looks at the period 2005–2012,

⁴ As an example of how double-counting can occur, silicon usage in PV modules is sometimes expressed in units of grams per watt. However, silicon usage per watt can decrease either because of a reduced silicon need *per wafer*, or because of an efficiency improvement. Efficiency of PV modules improved at the same time that silicon needs *per wafer* decreased. As a result, silicon usage *per watt* has changed more than silicon usage *per wafer* has. If the full cost reduction benefit of higher efficiency were already counted elsewhere, then the silicon-usage benefit of higher efficiency will be double-counted.

Double-counting is less likely to occur if the analysis of cost changes starts with an equation for cost. In constructing this equation, the grams-per-watt units of silicon usage would have to be reconciled with the watts units of efficiency $\times$ wafer area $\times$ the solar constant to recover the correct units for cost. Once this reconciling of units is done, the cost change equations that are derived from the cost equation will automatically avoid double-counting as well.

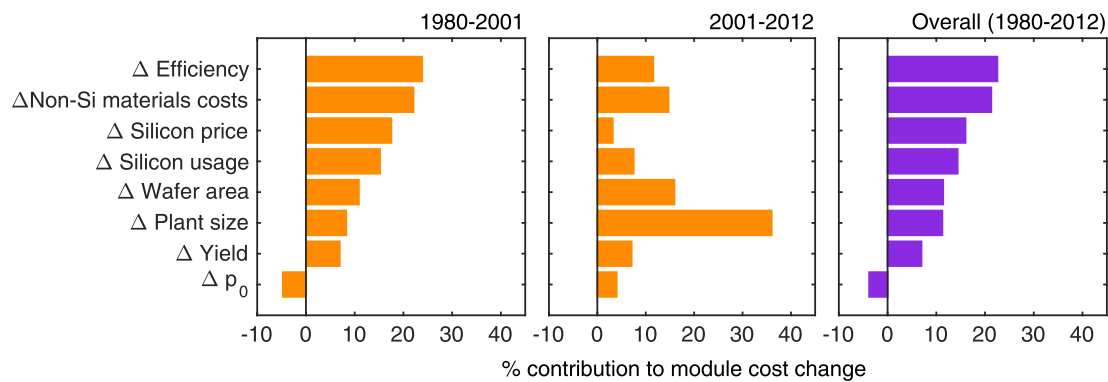


Fig. 3. Contribution of the low-level mechanisms to module cost decline in 1980–2001 (left), 2001–2012 (middle), and 1980–2012 (right). Mechanisms are listed in the order of decreasing contribution for the 1980–2001 period.

Table 3

Contribution of the low-level mechanisms to module cost decline in 1980–2001 (left), 2001–2012 (middle), and 1980–2012 (right). Mechanisms are listed in the order of decreasing contribution for the 1980–2001 period. Costs are in 2015 USD.

Cost change due to:	1980–2001		2001–2012		1980–2012	
	Δ\$/W	%	Δ\$/W	%	Δ\$/W	%
Δ Efficiency	–5.96	24%	–0.35	12%	–6.30	23%
Δ Non-Si materials costs	–5.51	22%	–0.44	15%	–5.95	21%
Δ Silicon price	–4.38	18%	–0.10	3%	–4.47	16%
Δ Silicon usage	–3.80	15%	–0.23	8%	–4.02	14%
Δ Wafer area	–2.71	11%	–0.48	16%	–3.19	11%
Δ Plant size	–2.07	8%	–1.08	36%	–3.15	11%
Δ Yield	–1.73	7%	–0.21	7%	–1.95	7%
Δ p₀	1.18	–5%	–0.12	4%	1.06	–4%
Change in module cost	–24.99	100%	–3.00	100%	–27.99	100%

we find a similar contribution by efficiency increases, and a smaller contribution by silicon price and silicon usage reductions, with plant size instead making the single largest contribution.

4.2. High-level mechanisms of cost reduction

Low-level cost reductions can be attributed to various ‘high-level’ mechanisms such as research and development, learning-by-doing, and economies of scale. Estimating the contributions of these mechanisms is useful because they align more closely with the policy levers often used to drive down cost. To estimate how much each contributed to cost reduction

in PV modules (Fig. 4), we categorize each low-level mechanism according to the high-level mechanisms responsible for it. In Section S4 of the Supporting Information we perform a sensitivity analysis to test the effect of these assumptions. Our conclusions about the relative importance of different high-level mechanisms are robust to various schemes for relating low-level and high-level mechanisms (see Fig. S20).

Changes that require a lab setting or a nonroutine production activity (e.g. experimental production line) are labeled as being caused by research and development (Pisano, 1996; Rosenberg, 1982). R&D can result in improvements to either the manufacturing process or the technology being produced. We consider an improvement to have been made by learning-by-doing (LBD) if it was achieved as a result of repeated routine manufacturing activity and if it was incremental in nature (Pisano, 1996). Cost changes that result from increases to the module manufacturing plant scale, and from volume purchases of materials or scale economies in materials supplier industries, we categorize as economies of scale (EOS).

Based on this we categorize improvements to module efficiency, wafer area, and silicon usage under R&D. Improvements to cell efficiency were largely achieved by R&D done at national labs, universities, and companies (NREL, 2018; Green, 2005; Goetzberger et al., 2003; Bruton, 2002; Rohatgi, 2003). Closing the gap between cell and module efficiencies also required R&D to improve module assembly processes such as encapsulation (Green, 2005). Larger wafer area was achieved through R&D on single crystal growth and multicrystalline ingot casting (Christensen, 1985). Wafer thickness and silicon utilization improved through manufacturing techniques such as wire-sawing that were improved through R&D (Luque, 1987; Green, 2000; Kazmerski, 2006; Goetzberger et al., 2003). LBD may have been an additional driver of wafer area and silicon usage, which are

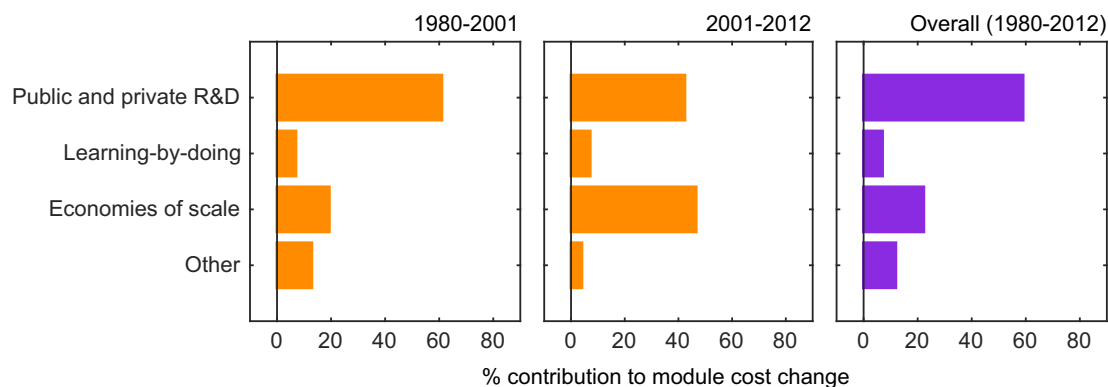


Fig. 4. Contribution of the high-level mechanisms to module cost decline in 1980–2001 (left), 2001–2012 (middle), and 1980–2012 (right). We categorize the changes that require a lab setting or a nonroutine production activity (e.g. experimental production line) as being caused by research and development (Pisano, 1996; Rosenberg, 1982). We consider an improvement to have been made by learning-by-doing if it was achieved as a result of repeated routine manufacturing activity and if it was incremental in nature (Pisano, 1996). We categorize the changes that result from increases to the scale of the module manufacturing plant and from volume purchases of materials as economies of scale.

affected by the efficiency of the manufacturing process.

Yield likely improved mainly through LBD, as advances in quality control of wafers and cells reduced rejects, and automation reduced excess handling (Swanson, 2006). R&D may have played a role, though we expect that improving yield mainly involved repeated routine manufacturing activity.

The increasing size of module plants brought about economies of scale (EOS), as manufacturers simultaneously prepared for higher demand (Nemet, 2006) and looked for better access to capital (Goodrich et al., 2013b). Larger plants realized cost savings from spreading out the costs of shared infrastructure across greater output and from physical or geometric scaling relationships (Bruni, 1964; Silberston, 1972). In our model we assume all non-material costs realize scale benefits so that increasing plant sizes lowers the cost per watt of labor, maintenance, capital equipment, and electricity.

Silicon prices were driven by different developments over time. Until the mid-2000s, demand for silicon by the PV industry was met mainly with electronic grade silicon rejects from the semiconductor industry (Woditsch and Koch, 2002; Sarti and Einhaus, 2002). The availability of silicon was mainly a positive externality of semiconductor production. In this period knowledge spillover was important (Swanson, 2006). The semiconductor industry developed know-how for producing single crystal silicon ingots and slicing wafers that benefited the PV industry. We therefore categorize silicon price's high-level mechanism as 'other' for 1980–2001. The PV industry surpassed the semiconductor industry in silicon demand around 2006 (Fisher et al., 2012), leading to a price spike and supply shortage. To meet the demand, polysilicon producers rapidly scaled up their capacities (Fu et al., 2015). For 2001–2012 we therefore choose EOS as the main high-level mechanism for decreasing silicon price.

Decreases to non-silicon materials costs were important sources of cost reduction in both time periods. Non-silicon materials costs can be decomposed into material usage (mass/area) times material price (dollars/mass). We propose that R&D helped reduce materials usage through new module designs (Green, 2005; ITRPV, 2018), while EOS led to decreasing prices due to volume purchases or scale economies in materials supplier industries (Silberston, 1972; Goodrich et al., 2013b). We assign decreases in non-silicon materials costs equally to R&D and EOS, and explore the effect of other assignments in Fig. S20.

Finally we choose 'other' as the high-level mechanism for the change in p_0 . p_0 includes different types of costs (electricity, labor, maintenance, and depreciation) and multiple high-level mechanisms can affect it. For example, the labor costs component of p_0 can be reduced through LBD. We do not have the information to break down p_0 , and therefore cannot quantify the high-level mechanisms governing it. However, this issue does not affect the results much since the module cost change due to the change in p_0 has been small.

Adding the effects of low-level mechanisms, we estimate the percent changes in module cost due to R&D, LBD, and EOS in the two time periods (Fig. 4). In the Supporting Information, we consider other plausible categorizations of low-level mechanisms and show the maximum and minimum values achieved with these alternate categorizations. Fig. S20 shows two alternative cases that include recategorizing c , A , and v . These two alternative categorizations are meant to reflect the lowest and highest values R&D, EOS, and LBD can take, so that robust conclusions can be drawn.

Our estimates show a strong impact from R&D. R&D played a dominant role in the first period, improving multiple cost equation variables that reduced cost significantly (Table 3). R&D's impact is high in the second period as well. LBD is estimated to have had a small impact, though we show in Fig. S20 that LBD's impact could be higher to the extent that it contributed to improvements in wafer area and silicon usage.

Scale economies changed from being a minor contributor in the first period to a significant one in the second. In the first period, though the change in plant size was responsible for significant cost reductions in absolute terms, cost reductions overall were dominated by R&D-driven

efficiency and material usage improvements. These improvements slowed down and plant size grew more in percentage terms in the second period (Table 1), permitting EOS to contribute a larger share.

Private R&D, learning-by-doing, and scale economies were all catalyzed by market-stimulating policies (e.g. through feed-in-tariffs, renewable portfolio standards). It is challenging to separate the effects of private and public R&D, though data suggests that private and public R&D expenditures have been similar in magnitude (McCrone et al., 2016, 2017; Nemet and Kammen, 2007). (See Section S5.) As a rough estimate, if 50% of R&D expenditures were from private R&D, and if private R&D expenditures have the same tendency to reduce cost as public R&D expenditures, then market-stimulating policies would have contributed about 60% of cost reduction over both periods. These results are shown in Fig. 5, where uncertainty bars reflect the range in contribution from market-stimulating policies that would result under the alternate assignments of low-level mechanisms to high-level mechanisms given in Fig. S20. Even without relying on funding levels as a proxy, there is evidence of private R&D's impact on variables such as efficiency, wafer area, silicon, and non-silicon materials usage, all of which developed in conjunction with changes to the manufacturing process (NREL, 2018; Green, 2005; Goetzberger et al., 2003; Christensen, 1985; Kazmerski, 2006; Bruton, 2002; Rohatgi, 2003; ITRPV, 2018). Moreover, if we consider the unlikely scenario in which private R&D is assumed to have no effect on cost at all, market-stimulating policies would still be estimated to contribute roughly a third of the cost decline through EOS and LBD (Fig. 3). These results demonstrate that PV is an example of a technology where market-stimulating policies played a significant role in encouraging innovative activity and driving down costs (Bettencourt et al., 2013; Huenteler et al., 2016).

Some cost reductions came from improvements made outside of the PV industry, and were not stimulated by PV market expansion. We make a rough estimate of the upper bound on these effects using the variables that were affected by outside developments. p_0 includes equipment costs in module manufacturing, and some equipment improvements took place in other industries and were transmitted to module manufacturing as knowledge spillovers. The change in wafer area may have come from improvements in the semiconductor industry. As noted earlier the price of silicon was also affected by developments in the semiconductor industry. Summing the cost reductions from these three variables would give us an estimate of 23% of PV's cost reduction coming from developments outside the PV industry in 1980–2012.

4.3. Prospective cost reductions

Considering the various low-level and high-level causes of historical cost declines, how effective would different strategies be at reducing module costs going forward? To gain insight we perform two simple analyses to assess how influential each low-level and high-level mechanism is for reducing costs under our model. In the first analysis, each cost equation variable is changed one-at-a-time, starting from its 2012 value, in a direction that reduces cost. With the exceptions of yield and plant size we change each variable by the same percentage of $\pm 25\%$ to see how much cost reduction results. While different variables are not equally likely to realize a given percentage improvement, this approach lets us see how strongly each variable influences cost. To avoid using an unphysical yield above 100% instead of raising it by 25% we set yield equal to 100%. Plant size historically grew by very large factors far exceeding 25% – 13-fold in 1981–2001, and 75-fold in 2001–2012, and a 25% increase would not be expected to achieve much cost reduction. Instead we consider a 3-fold and a 10-fold increase in plant size. A 3-fold increase would result in plant sizes of about 3 GW/year, slightly larger than the size of the largest plants in China in 2012 (Pillai, 2015). The results are shown in the top panel of Fig. 6. The most influential variables are efficiency, plant size, and non-silicon materials costs. Plant size contributes substantially, though as plants become larger it may become more difficult to increase plant sizes by factors large

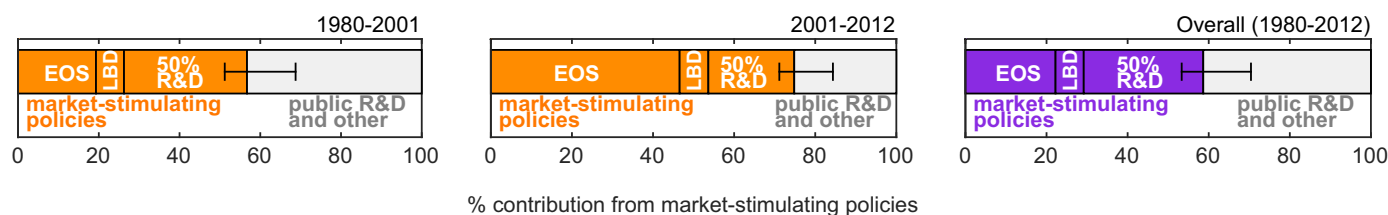


Fig. 5. Contribution of market-stimulating policies (e.g. feed-in-tariffs, renewable portfolio standards) to module cost reduction in 1980–2001 (left), 2001–2012 (middle), and 1980–2012 (right). R&D = research and development, LBD = learning-by-doing, EOS = economies of scale, Other = other mechanisms such as spillovers. Scale economies, learning-by-doing, and private R&D were all catalyzed by market-stimulating policies. Our data does not let us separate the effects of private and public R&D. To accommodate this, we add 50% of total R&D contributions to the contributions of scale economies and learning-by-doing. This effectively assumes that 50% of R&D improvements came from private R&D (roughly commensurate with the share of R&D expenditures in clean energy (McCrone et al., 2016, McCrone et al., 2017; Nemet and Kammen, 2007)). (See Section S5 for the approach used to obtain this estimate.) Uncertainty bars show the total contribution from market-stimulating policies that would result under the alternate assignments of low-level mechanisms to high-level mechanisms shown in Fig. S20, without accounting for uncertainty in the private R&D estimate.

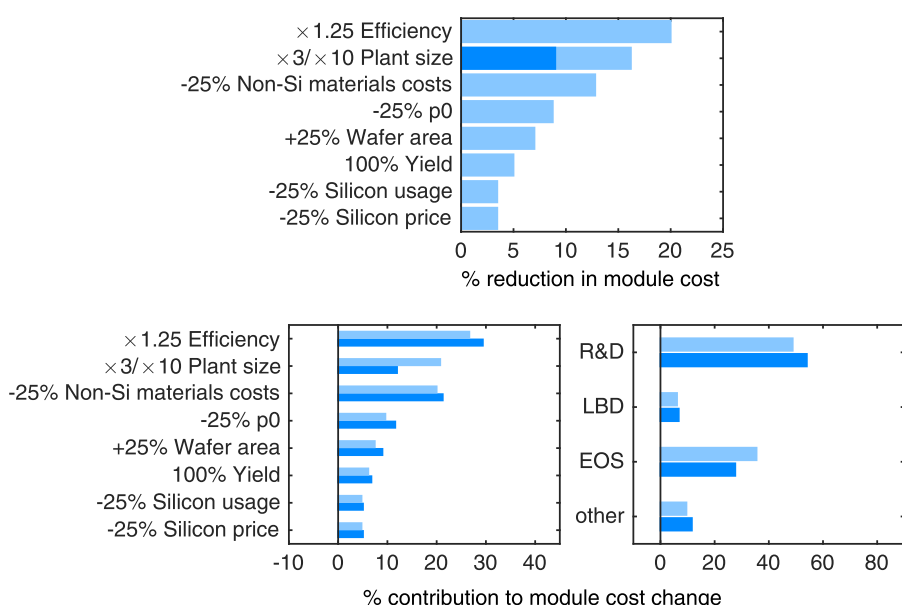


Fig. 6. Prospective cost reductions. Top: Module cost reductions from one-at-a-time changes to cost equation variables. The reduction in module cost below the 2012 value is shown for each one-at-a-time variable change. Most variables are adjusted up or down by 25%, in a direction that reduces cost. Yield was changed from 95% to 100%. Plant size is increased by a factor of 3 (dark blue) and a factor of 10 (light blue). Bottom: Percent contributions of low-level mechanisms (left) and high-level mechanisms (right) to module cost reduction below the 2012 value in a scenario where several cost equation variables are changed simultaneously by the amounts shown. Separate bars are shown for the cases where plant size increases by a factor of 3 (dark blue) and a factor of 10 (light blue). R&D = research and development, LBD = learning-by-doing, EOS = economies of scale, Other = other mechanisms such as spillovers.

enough to realize significant further gains.

High-level mechanisms often encompass improvements in several variables at the same time, and therefore we also model the simultaneous occurrence of multiple low-level mechanisms. To assess the cost change contribution of each low-level mechanism, we need to use the cost change equations developed in Section 3. (This is in contrast to the one-at-a-time changes to variables described above, whose contributions to cost change can be computed by comparing total costs at each snapshot in time.) In this second analysis we alter all variables simultaneously in the amounts given above. We then group the cost changes from these variables by high-level mechanism in the same way as in the second period of our historical analysis. The results⁵ are shown in the bottom row of Fig. 6. Changing all variables at once, with a 10-fold increase in plant size, module costs decline to 44% of the 2012 value. As was observed historically, combined public and private R&D accounts for most of the decline in this scenario.

Our model can be used as a tool to perform prospective analyses to guide future engineering and policy efforts. The above analysis shows one scenario in which high-level mechanisms drive several low-level

mechanisms. In this scenario, summing the percent contributions of learning-by-doing and scale economies, in the case of a 10X plant size increase, about 40% of total cost reduction results from mechanisms related to market-expansion (or roughly 65% if we assign 50% of future R&D-related improvements to the private sector, commensurate with the past share of private R&D funding (McCrone et al., 2016, 2017; Nemet and Kammen, 2007) as explained in Section S5).

Several scenarios could be developed to prospectively investigate the effects of low-level mechanisms, public R&D funding strategies, market-stimulating policies, private R&D, and other firm-level strategy. Expert elicitation could be used to assess the potential for changing variables in the cost equation. (The quantitative framework can enable the elicitation to focus on detailed variables that technical experts are familiar with, and may therefore help to elicit more reliable information from experts than has traditionally been achieved (Hultman and Koomey, 2007; Morgan, 2014).) Using this information, an exploration of the effects of different combinations of high- and low-level mechanisms could inform decisions in the public and private sectors. This approach could be applied not only to PV modules, but PV systems, energy storage, and other technologies, as well as performance metrics other than cost.

5. Conclusions and policy implications

PV module costs declined dramatically in the last forty years. To identify the reasons, we advance an analysis method that bridges top-

⁵ In the 10X plant size case, plant size and non-Si materials costs swap places in the ranking of their effects. This can happen since changing all variables at once has different effects on cost than changing variables one-at-a-time and then aggregating their effects.

down, high-level analyses of technology cost evolution with bottom-up engineering models. The method begins with a cost equation relating a technology's costs to a set of underlying variables, e.g. technical performance characteristics such as efficiency or material usage. Cost change equations are then derived to estimate each variable's contribution. These variables explain cost changes at a 'low level', and we term changes to these variables low-level mechanisms of cost reduction. Other processes like R&D, learning-by-doing, and scale economies may encompass several low-level mechanisms and drive down costs as a whole. We term these high-level causes of cost reduction.

We find that cost reductions in PV modules were fairly evenly distributed across a number of low-level mechanisms, which may help explain why this technology experienced relatively steady cost reductions over the past three decades. We estimate that changes to efficiency contributed 23% to the cost reduction from 1980 to 2012, non-silicon materials costs 21%, silicon price 16%, silicon usage 14%, wafer area 11%, plant size 11%, and yield 7%.

Several high-level mechanisms were important. These include public and private R&D and economies of scale, which contributed an estimated 59% and 22%, respectively, of the cost decline between 1980 and 2012. Learning-by-doing, defined here as incremental and resulting from a repeated, routine manufacturing process (e.g. not requiring an experimental production line), contributed 7% of the cost decline during this period.

Looking across both the 1980–2001 and the 2001–2012 periods, our findings suggest that the key drivers of decreasing costs have been changing over time. Economies of scale in particular have had a greater impact more recently, and likely offer an avenue for further cost reductions. Notably, the typical 2012 plant size in our data set has been surpassed by several new Chinese plants with typical sizes of 1–2 GW/year (Goodrich et al., 2013b). However there may be a limit to how much plant sizes will grow, and savings from economies of scale may be exhausted over time.

R&D, both public and private, was a key driver of module cost reduction historically and can be valuable going forward in improving module efficiency and reducing materials use. Improvements to module efficiency in particular would help cut the per-watt cost of all cost components of PV modules (as well as PV systems). Variables that might face limitations in the short term are manufacturing yield, which is already close to 100%, and wafer area, which is constrained by yield and efficiency considerations.

Market-stimulating policies have played a central role in driving down the costs of PV modules, with private R&D, economies of scale, and learning-by-doing together contributing an estimated 60% of the cost decline in PV modules between 1980 and 2012. This finding contributes to an ongoing debate on the effectiveness of market-expansion policies, as complements to publicly-funded R&D (Hoppmann et al., 2013). In the case of PV, our analysis shows that private sector activity, which was incentivized by government policies in various nations (Trancik et al., 2015), was critically important for driving down costs. Additionally, our findings support the importance of public R&D to complement private sector activity, which may focus more on refinements to technology and manufacturing rather than the more major innovations needed as the limits to incremental improvements are reached (Hoppmann et al., 2013; Zheng and Kammen, 2014). These results add insight to earlier findings from correlational analyses on the importance of market expansion in driving energy patenting (as a proxy for innovative activity) in recent years (Bettencourt et al., 2013), with public R&D having played the dominant role in the 70s and 80s.

Looking forward, market-stimulating policies can continue to support cost declines, through a virtuous, mutually-reinforcing cycle of technology improvement and emissions reductions (Trancik et al., 2015). However, our results suggest that these policies should be complemented by public R&D to reduce the risks of exhausting the improvement potential of current-generation, silicon-based module technology, by exploring devices based on other abundant semiconductor materials with the potential for lower processing costs (Kavlak et al., 2015).

In this work we provide a reconstruction of trends in PV modules using three representative points in time, drawing from a variety of sources. It is important to note that the data used in this work has uncertainties, for

example due to the incomplete sampling of firms by the publications we obtained our data from. Nevertheless we provide a framework for modeling costs and cost changes that can be populated with the best available data.

This analysis can be helpful in guiding engineering efforts, the formulation of energy and climate policy, and research investments by government and the private sector. The method outlined here can be applied to many technologies to understand the reasons for past changes in cost or other performance metrics, and to explore promising opportunities for future improvement.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.enpol.2018.08.015.

References

- Applied Materials, 2011. Wafer Wire Sawing Economics and Total Cost of Ownership Optimization, White Paper. URL <http://www.appliedmaterials.com/sites/default/files/Wire-Sawing-Economics-and-COO-Whitepaper_en.pdf>. (Accessed on 9 October 2017).
- Bettencourt, L.M.A., Trancik, J.E., Kaur, J., 2013. Determinants of the pace of global innovation in energy technologies. *PLoS One* 8, e67864.
- Braff, W.A., Mueller, J.M., Trancik, J.E., 2016. Value of storage technologies for wind and solar energy. *Nat. Clim. Change* 6, 964–970.
- Bruni, L., 1964. Internal economies of scale with a given technique. *J. Indus. Econ.* 12, 175–190.
- Bruton, T.M., 2002. General trends about photovoltaics based on crystalline silicon. *Solar Energy Materials and Solar Cells* 72, 3–10. EMRS 2001 Symposium E: Crystalline Silicon for Solar Cells.
- Candelise, C., Winkler, M., Gross, R.J., 2013. The dynamics of solar PV costs and prices as a challenge for technology forecasting. *Renew. Sustain. Energy Rev.* 26, 96–107.
- Carbajales-Dale, M., Barnhart, C.J., Brandt, A.R., Benson, S.M., 2014. A better currency for investing in a sustainable future. *Nat. Clim. Change* 4, 524–527.
- Centrotherm Photovoltaics, 2010. PV Panels for the US: Make - not Buy! URL <http://www.gaccsouth.com/fileadmin/ahk_atlanta/Bilder_Houston/Bilder/Texas_Renewables/05_Centrotherm_Faller.ppt_ohne_extra_slides.pdf_web.pdf>. (Accessed on 30 April 2018).
- Christensen, E., 1985. Electricity from Photovoltaic Solar Cells. Flat-Plate Solar Array Project of the U.S. Department of Energy's National Photovoltaics Program. 10 Years of Progress. Technical Report. US DOE, JPL and NASA.
- Costello, D., Rappaport, P., 1980. The technological and economic development of photovoltaics. *Ann. Rev. Energy* 5, 335–356.
- del Canizo, C., del Coso, G., Sinke, W.C., 2009. Crystalline silicon solar module technology: Towards the 1 € per watt-peak goal. *Prog. Photovolt. Res. Appl.* 17, 199–209.
- de La Tour, A., Glachant, M., Meniere, Y., 2013. Predicting the costs of photovoltaic solar modules in 2020 using experience curve models. *Energy* 62, 341–348.
- Duke, R., Kammen, D.M., 1999. The economics of energy market transformation programs. *Energy J.* 20, 15–64.
- Feldman, D., Barbose, G.L., Margolis, R., James, T., Weaver, S., Darghouth, N.R., Fu, R., Davidson, C., Booth, S., Wiser, R.H., 2014. Photovoltaic System Pricing Trends: Historical, Recent, and Near-Term Projections - 2014 Edition. Technical Report. NREL.
- Fisher, G., Seacrist, M.R., Standley, R.W., 2012. Silicon crystal growth and wafer technologies. *Proc. IEEE* 100, 1454–1474.
- Fraunhofer Institute, 2015. Current and Future Cost of Photovoltaics. Long-term Scenarios for Market Development, System Prices and LCOE of Utility-Scale PV Systems. Study on behalf of Agora Energiewende. Technical Report. Freiburg, Germany. URL <http://www.fvee.de/fileadmin/publikationen/weitere_publikationen/15_AgoraEnergiewende-ISE_Current_and_Future_Cost_of_PV.pdf>. (Accessed on 9 October 2017).
- Fraunhofer Institute, 2017. Photovoltaics Report. Technical Report. Freiburg, Germany. URL <<https://www.ise.fraunhofer.de/content/dam/ise/de/documents/publications/studies/Photovoltaics-Report.pdf>>. (Accessed on 9 October 2017).
- Fu, R., James, T., Woodhouse, M., 2015. Economic measurements of polysilicon for the photovoltaic industry: Market competition and manufacturing competitiveness. *IEEE J. Photovolt.* 5, 515–524.
- Funk, J.L., 2013. What drives exponential improvements? *Calif. Manag. Rev.* 55, 134–152.
- Ghannam, M., Sivoththaman, S., Poortmans, J., Szlufcik, J., Nijs, J., Mertens, R., Overstraeten, R.V., 1997. Trends in industrial silicon solar cell processes. *Solar Energy* 59, 101–110. Selected Proceedings of ISES 1995: Solar World Congress. Part III.
- Goetzberger, A., Hebling, C., Schock, H.W., 2003. Photovoltaic materials, history, status

- and outlook. *Mater. Sci. Eng.: R Rep.* 40, 1–46.
- Goodrich, A.C., Hacke, P., Wang, Q., Sopori, B., Margolis, R., James, T.L., Woodhouse, M., 2013a. A wafer-based monocrystalline silicon photovoltaics road map: Utilizing known technology improvement opportunities for further reductions in manufacturing costs. *Sol. Energy Mater. Sol. Cells* 114, 110–135.
- Goodrich, A.C., Powell, D.M., James, T.L., Woodhouse, M., Buonassisi, T., 2013b. Assessing the drivers of regional trends in solar photovoltaic manufacturing. *Energy Environ. Sci.* 6, 2811–2821.
- Green, M.A., 2000. The future of crystalline silicon solar cells. *Prog. Photovolt. Res. Appl.* 8, 127–139.
- Green, M.A., 2005. Silicon photovoltaic modules: a brief history of the first 50 years. *Prog. Photovolt. Res. Appl.* 13, 447–455.
- Grubler, A., Nakicenovic, N., Victor, D.G., 1999. Dynamics of energy technologies and global change. *Energy Policy* 27, 247–280.
- Hammond, A.L., 1977. Photovoltaics: The semiconductor revolution comes to solar. *Science* 197, 445–447.
- Hertwich, E.G., Gibon, T., Bouman, E.A., Arvesen, A., Suh, S., Heath, G.A., Bergesen, J.D., Ramirez, A., Vega, M.I., Shi, L., 2015. Integrated life-cycle assessment of electricity-supply scenarios confirms global environmental benefit of low-carbon technologies. *Proc. Natl. Acad. Sci. USA* 112, 6277–6282.
- Hoppmann, J., Peters, M., Schneider, M., Hoffmann, V.H., 2013. The two faces of market support—how deployment policies affect technological exploration and exploitation in the solar photovoltaic industry. *Res. Policy* 42, 989–1003.
- Huenteler, J., Schmidt, T.S., Ossenbrink, J., Hoffmann, V.H., 2016. Technology life-cycles in the energy sector - technological characteristics and the role of deployment for innovation. *Technol. Forecast. Soc. Change* 104, 102–121.
- Hultman, N.E., Koomey, J.G., 2007. The risk of surprise in energy technology costs. *Environ. Res. Lett.* 2, 034002.
- ITRPV, 2012. International Technology Roadmap for Photovoltaics Results 2011, Third Edition. Technical Report.
- ITRPV, 2018. International Technology Roadmap for Photovoltaics Results 2017, Ninth Edition. Technical Report.
- Jones-Albertus, R., Feldman, D., Fu, R., Horowitz, K., Woodhouse, M., 2016. Technology advances needed for photovoltaics to achieve widespread grid price parity. *Prog. Photovolt. Res. Appl.* 24, 1272–1283 PIP-15-239.R1.
- Kavlak, G., McEnerney, J., Jaffe, R.L., Trancik, J.E., 2015. Metal production requirements for rapid photovoltaics deployment. *Energy Environ. Sci.* 8, 1651–1659.
- Kazmerski, L.L., 2006. Solar photovoltaics R&D at the tipping point: A 2005 technology overview. *J. Electron Spectrosc. Relat. Phenom.* 150, 105–135 Science and Spectroscopy of Environmentally Important Interfaces.
- Koh, H., Magee, C.L., 2008. A functional approach for studying technological progress: extension to energy technology. *Technol. Forecast. Soc. Change* 75, 735–758.
- Little, R.G., Nowlan, M.J., 1997. Crystalline silicon photovoltaics: the hurdle for thin films. *Prog. Photovolt. Res. Appl.* 5, 309–315.
- Luque, A., 1987. Photovoltaics in 1986: Routes to low cost, in: Goetzberger, A., Palz, W., Willeke, G. (Eds.), Seventh E.C. In: Proceedings of the Photovoltaic Solar Energy Conference, Springer Netherlands, Dordrecht. pp. 9–18.
- Maycock, P.D., 1995. International photovoltaic markets, developments and trends forecast to 2010. *Renew. Energy* 6, 469–475.
- Maycock, P.D., 1997. Cost reduction in PV manufacturing impact on grid-connected and building-integrated markets. *Sol. Energy Mater. Sol. Cells* 47, 37–45.
- McCrone, A., Moslener, U., d'Estais, F., Gruuning, C., Louw, A., Boyle, R., Strahan, D., Collins, B., Stopforth, K., Becker, L., 2017. Global Trends in Renewable Energy Investment 2017. Technical Report. UNEP and Bloomberg New Energy Finance.
- McCrone, A., Moslener, U., d'Estais, F., Usher, E., Gruuning, C., Byrne, J., Mills, L., Strahan, D., Boyle, R., Collins, B., Stopforth, K., Becker, L., 2016. Global Trends in Renewable Energy Investment 2016. Technical Report. UNEP and Bloomberg New Energy Finance.
- McDonald, A., Schratzenholzer, L., 2001. Learning rates for energy technologies. *Energy Policy* 29, 255–261.
- McEnerney, J., Farmer, J.D., Trancik, J.E., 2011. Historical costs of coal-fired electricity and implications for the future. *Energy Policy* 39, 3042–3054.
- Mints, P., 2015. Historical PV data. Personal Commun.
- Mitchell, R., Witt, C., King, R., Ruby, D., 2002. PVMaT advances in the photovoltaic industry and the focus of future PV manufacturing R&D. In: Conference Record of the 29th IEEE Photovoltaic Specialists Conference, pp. 1444–1447.
- Morgan, M.G., 2014. Use (and abuse) of expert elicitation in support of decision making for public policy. *Proc. Natl. Acad. Sci.* 111, 7176–7184.
- Nagy, B., Farmer, J.D., Bui, Q.M., Trancik, J.E., 2013. Statistical basis for predicting technological progress. *PLoS One* 8, e52669.
- National Renewable Energy Laboratory, 2018. Research Cell Record Efficiency Chart. URL <<https://www.nrel.gov/pv/assets/images/efficiency-chart.png>>. (Accessed on 19 May 2018).
- Needell, Z.A., McEnerney, J., Chang, M.T., Trancik, J.E., 2016. Potential for widespread electrification of personal vehicle travel in the United States. *Nat. Energy* 1, 16112.
- Nemet, G.F., 2006. Beyond the learning curve: factors influencing cost reductions in photovoltaics. *Energy Policy* 34, 3218–3232.
- Nemet, G.F., Kammen, D.M., 2007. U.S. energy research and development: Declining investment, increasing need, and the feasibility of expansion. *Energy Policy* 35, 746–755.
- Pillai, U., 2015. Drivers of cost reduction in solar photovoltaics. *Energy Econ.* 50, 286–293.
- Pillai, U., Cruz, K., 2013. Source of Cost Reduction in Solar Photovoltaics. URL <<http://ssrn.com/abstract=2259155>>. (Accessed on 9 October 2017).
- Pillai, U., McLaughlin, J., 2013. A model of competition in the solar panel industry. *Energy Econ.* 40, 32–39.
- Powell, D.M., Winkler, M.T., Choi, H.J., Simmons, C.B., Needleman, D.B., Buonassisi, T., 2012. Crystalline silicon photovoltaics: a cost analysis framework for determining technology pathways to reach baseload electricity costs. *Energy Environ. Sci.* 5, 5874–5883. URL <<http://pv.mit.edu/TMA>>.
- Pisano, G.P., 1996. Learning-before-doing in the development of new process technology. *Res. Policy* 25, 1097–1119.
- Powell, D.M., Winkler, M.T., Goodrich, A., Buonassisi, T., 2013. Modeling the cost and minimum sustainable price of crystalline silicon photovoltaic manufacturing in the united states. *IEEE J. Photovolt.* 3, 662–668. URL <<http://pv.mit.edu/TMA>>.
- Ravi, K.V., 2013. Silicon solar cell manufacturing costs - Where have we been and what is needed going forward. Present. MIT PV Class 6.262.
- Reichelstein, S., Sahoo, A., 2014. Cost and Price Dynamics of Solar PV Modules. Working Paper No. 3069. Graduate School of Business, Steyer-Taylor Center for Energy Policy and Finance, Stanford University.
- Rohatgi, A., 2003. Road to cost-effective crystalline silicon photovoltaics, In: Proceedings of 3rd World Conference on Photovoltaic Energy Conversion, pp. A29–A34.
- Rosenberg, N., 1982. Inside the Black Box: Technology and Economics. Cambridge University Press, Cambridge, UK.
- Rubin, E.S., Azevedo, I.M., Jaramillo, P., Yeh, S., 2015. A review of learning rates for electricity supply technologies. *Energy Policy* 86, 198–218.
- Sagar, A.D., van der Zwaan, B., 2006. Technological innovation in the energy sector: R&D, deployment, and learning-by-doing. *Energy Policy* 34, 2601–2608.
- Sarti, D., Einhaus, R., 2002. Silicon feedstock for the multi-crystalline photovoltaic industry. *Sol. Energy Mater. Sol. Cells* 72, 27–40.
- Silberston, A., 1972. Economies of scale in theory and practice. *Econ. J.* 82, 369–391.
- Swanson, R.M., 2006. A vision for crystalline silicon photovoltaics. *Prog. Photovolt. Res. Appl.* 14, 443–453.
- Swanson, R.M., 2011. The Silicon Photovoltaics Roadmap. URL <<http://energyseminar.stanford.edu/sites/all/files/eventpdf/11-11-14%20Swanson.pdf>>. (Accessed on 9 October 2017).
- Symko-Davies, M., Mitchell, R.L., Witt, C.E., Thomas, H.P., King, R., Ruby, D., 2000. Decade of PV industry R&D advances in silicon module manufacturing, In: Conference Record of the 28th IEEE Photovoltaic Specialists Conference - 2000 (Cat. No.00CH37036), pp. 1460–1463.
- Trancik, J.E., 2014. Renewable energy: Back the renewables boom. *Nature* 507, 300–302.
- Trancik, J.E., Brown, P.R., Jean, J., Kavlak, G., Klemun, M.M., Edwards, M.R., McEnerney, J., Miotti, M., Mueller, J.M., Needell, Z.A., 2015. Technology Improvement and Emissions Reductions as Mutually Reinforcing Efforts: Observations from the Global Development of Solar and Wind Energy. Technical Report. Institute for Data, Systems, and Society, MIT. URL <<http://energy.mit.edu/publication/technology-improvement-and-emissions-reductions-as-mutually-reinforcing-efforts/>>. (Accessed on 9 October 2017).
- Trancik, J.E., Cross-Call, D., 2013. Energy technologies evaluated against climate targets using a cost and carbon trade-off curve. *Environ. Sci. Technol.* 47, 6673–6680.
- U.S. Department of Energy, 2012. SunShot Vision Study. Technical Report. U.S. Department of Energy. Washington, DC.
- U.S. Geological Survey, 2012. Silicon Minerals Yearbook. Technical Report. U.S. Geological Survey. Reston, VA.
- van der Zwaan, B., Rabl, A., 2004. The learning potential of photovoltaics: implications for energy policy. *Energy Policy* 32, 1545–1554.
- Williams, B.F., 1980. Do photovoltaics have a future? *J. Eng. Gas Turbines Power* 102, 495–497.
- Woditsch, P., Koch, W., 2002. Solar grade silicon feedstock supply for PV industry. *Solar Energy Materials and Solar Cells* 72, 11–26. EMRS 2001 Symposium E: Crystalline Silicon for Solar Cells.
- Woodhouse, M., Goodrich, A., Margolis, R., James, T., Dhare, R., Gessert, T., Barnes, T., Eggert, R., Albin, D., 2013a. Perspectives on the pathways for cadmium telluride photovoltaic module manufacturers to address expected increases in the price for tellurium. *Sol. Energy Mater. Sol. Cells* 115, 199–212.
- Woodhouse, M., Goodrich, A., Margolis, R., James, T.L., Lokanc, M., Eggert, R., 2013b. Supply-chain dynamics of tellurium, indium and gallium within the context of PV module manufacturing costs. *IEEE J. Photovolt.* 3, 833–837.
- Yu, C.F., van Sark, W.G.J.H.M., Alsema, E.A., 2011. Unraveling the photovoltaic technology learning curve by incorporation of input price changes and scale effects. *Renew. Sustain. Energy Rev.* 15, 324–337.
- Zheng, C., Kammen, D.M., 2014. An innovation-focused roadmap for a sustainable global photovoltaic industry. *Energy Policy* 67, 159–169.